

Emotional Impact on Investment and Insurance Preferences in Unit-Linked Insurance Plans

Peng-Yu Zeng

National Taiwan University

Hsuan-Yu Liu

National Taiwan University

Li-Yu Chen

National Taiwan University

Su-Ling Yeh

suling@ntu.edu.tw

National Taiwan University

Article

Keywords: Emotion, Unit-Link Insurance Plan, Investment, Insurance, Affect Infusion Model

Posted Date: September 18th, 2024

DOI: <https://doi.org/10.21203/rs.3.rs-4733981/v1>

License:  This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: No competing interests reported.

Abstract

A Unit-Link Insurance Plan (ULIP) is an emerging financial product that combines elements of both insurance and investment. Previous studies have found that positive and negative emotions predict investment and insurance products in opposite ways. This study aims to explore whether the influence of emotions on the willingness to purchase ULIPs is consistent with insurance products or aligns with investment products. This study invited policyholders to complete online questionnaires, using positive and negative emotional pictures and scenarios to manipulate emotions and observe their effects on the investment process of a ULIP product. The results revealed that both positive and negative emotions can impact the purchasing decisions of ULIP through distinct pathways. Positive and negative emotions directly contribute to increased cognitive assessment and purchasing intentions, with positive emotions demonstrating a stronger influence. Moreover, positive emotions indirectly enhance cognitive assessment and investment intentions by mediating an increase in the perceived performance and dividends. Our results support the affect infusion model and framing effect, suggesting that positive and negative emotions may lead participants to pay more attention to investment-oriented aspects and insurance-oriented aspects, respectively. These findings have implications for marketing strategies, product design, customer education, industry practices, and regulatory frameworks related to ULIPs.

Introduction

Unit-Linked Insurance Plans (ULIPs) have gained prominence due to their unique combination of investment and insurance benefits. Also known as “variable annuities” in the United States and “segregated fund policies” in Canada, ULIP is a hybrid life insurance designed to address both protection and investment aspects within the realm of financial products by incorporating investment components into insurance^{1,2}. In contrast to traditional life insurance, which offers guaranteed sums, ULIPs are typically linked to bonds or stocks, leading to dynamic investment performance. Consequently, policyholders not only secure their funds but also reap investment advantages. Due to these benefits, the global popularity of ULIPs has surged, with numerous Indian insurance companies actively promoting these products over conventional insurance offerings³. According to The Life Insurance Association of Taiwan (2023), considering premium revenue and debt, the income from ULIPs constitutes around 40% of total insurance policy income.

Customer emotions and the purchase of ULIP

Investment decisions often exhibit irrational behavior, heavily influenced by psychological factors, including emotions^{4,5}. In this context, emotions refer to experiences that include hedonic content and intensity⁶. The Affect Infusion Model (AIM) illustrates the processes through which emotions affect decision-making⁷. One process—termed the *affect-as-priming* process—suggests that emotional states indirectly influence judgment by affecting attention, encoding, retrieval, and associative processes during information retrieval from individual memory—our primary basis for judgment. Indeed, current affect can

alter selective attention processing, causing individuals to focus more on affect-congruent details than affect-incongruent ones⁸.

In the financial domain, AIM has been employed to explain risky financial behavior. Previous research found that emotions affect people's perception of risk in financial products, which in turn affects their willingness to buy, with positive emotional manipulation leading to a greater willingness to accept risk and negative emotions leading to a tendency to avoid risk⁹. In the process of valuing financial products, positive affect tends to encourage individuals' buying behavior, making them more prone to overvaluing the product. Conversely, negative affect is associated with perceiving a lower valuation of the product¹⁰. Chou et al. (2007)¹¹ found that compared with negative emotions, positive emotions incline individuals to perceive risky behaviors more favorably. Given that ULIP products are a risky commodity due to their net value fluctuating with market fluctuations, it can be inferred that emotions play a pivotal role in ULIP purchase decisions.

Divergent Effects of Emotions on Investment and Insurance Purchase Intentions

In the context of investments, positive emotions illuminate the favorable aspects of the financial market and boost the inclination to invest. Conversely, negative emotions dampen the desire to purchase financial products. The relationship between emotion and investment behavior can be elucidated through expectancy-value theories. According to these theories, having higher expectations of success instills greater confidence in achieving goals and thus encourages action¹². When we combine the *affect-as-priming* process in AIM and the expectancy-value theories, it is apparent individuals who experience positive emotions are more likely to perceive an optimistic future market outlook, which then leads to increased risk-taking behavior in the realm of finance¹³. For instance, hope can bring a sense of success, and promote people to take action towards expected goals¹⁴. Similarly, excitement, as a high-arousal positive cue, amplifies anticipated gains and encourages risk-taking behaviors¹⁵. In contrast, negative emotions foster a pessimistic outlook, prompting a more conservative attitude towards risk-taking. Pessimistic investors, for example, exhibit greater reluctance to invest for retirement compared to their optimistic counterparts¹⁶. Anxious college students expect bad outcomes from risky behavior and this leads to risk aversion^{17,18}. Narratives infused with fear induce a threat-based worldview, which reduces the willingness to engage in risky activities¹⁹.

The relationship between emotion and insurance purchasing differs from that observed in traditional investment. In the case of insurance, people often perceive it as a means of compensation and tend to find it more appealing when the product offers coverage of high-risk scenarios²⁰. A previous study found that Chinese citizens increased their health insurance purchases in response to heightened food safety risks²¹. Additionally, the emotion of loss motivates individuals to acquire life and casualty insurance, with uninsured individuals demonstrating an increased willingness to purchase insurance following a loss^{22,23}. Moreover, Yan et al. (2020)²⁴ found that fear, elicited through video content, increased private

insurance purchases. Although limited research directly explores the relationship between positive emotions and insurance purchases, Jaspersen and Aseervatham (2017)²⁵ found that positive emotions led people to believe in their safety in the aftermath of a loss and thus reduce the willingness to buy insurance.

Different framing effect of investment and insurance commodities

The contrasting predictions concerning the influence of emotion on purchase intentions for investment products, as opposed to insurance products, can be attributed to the framing effect. Unlike investment products that offer predictable financial returns, individuals cannot accurately predict accidents in the same way that insurance companies can^{26,27}. When it comes to low-probability, high-loss-risk accidents (e.g., death and fire) customers often struggle to make reliable assessments due to their limited real-life experiences²². Individuals are more willing to pay for airline travel insurance that covers negatively imaginable events such as terrorist acts compared to coverage for death from other causes²⁸. The imagination of negative events triggers the loss frame effect. However, in the investment field, positive emotions can mitigate perceived risk and emphasize profit, thereby promoting investment product purchases. Investment behavior emphasizes wealth accumulation, while insurance focuses on mitigating loss²⁹. Meta-analysis results indicate that positive and negative emotions are responsive to gain and loss framing, respectively³⁰. These contradictions imply the possibility of competition between insurance and investment products, especially when considering the influence of emotions.

Exploring emotion's impact on ULIP purchase intentions with investment and insurance characteristics

Prior studies have predominantly focused on decision-making processes for traditional investment and insurance products separately^{22,31}. However, ULIP products offer a unique opportunity to simultaneously evaluate both investment characteristics and insurance functions. In this study, we have developed an emotion induction procedure to investigate how emotions experienced by actual ULIP policyholders affect their willingness to invest in a hypothetical ULIP product profile within an investment scenario. Although previous research indicates that financial characteristics, such as the calculation of death benefits, determine whether ULIP products resemble insurance or investment³², individuals can adopt either an insurance or investment perspective during the purchase process. This study employs cognitive assessment and purchase intentions as measures to examine policyholders' preferences for ULIPs. Furthermore, while perceived performance and dividends are crucial pieces of information for ULIPs from the perspective of investment, they hold relatively less significance in the context of insurance objectives³³.

We hypothesize that if the purchase of ULIPs aligns more with traditional investment products, positive emotions will positively influence perceived performance, perceived dividends, cognitive assessment, and purchase intentions. Conversely, if the purchase of ULIPs aligns more with insurance products,

positive emotions will negatively predict cognitive assessment and purchase intentions, with no significant impact on the perception of investment information, such as perceived performance and perceived dividends (See Fig. 1).

H1: Positive emotion positively predicts perceived performance, perceived dividend, cognitive assessment, and purchase intention.

H1-Alternative: Positive emotion negatively predicts cognitive assessment and purchase intention, and it will not influence perceived performance and perceived dividend.

On the other hand, if the purchase of ULIP aligns with investment products, negative emotions will decrease performance, perceived dividends, cognitive assessment, and purchase intentions. Conversely, if the purchase of ULIPs aligns more with traditional insurance products, negative emotions will positively predict cognitive assessment and purchase intentions, with no impact on the perception of investment information, such as perceived performance and perceived dividends (See Fig. 2).

H2: Negative emotion negatively predicts perceived performance, perceived dividend, cognitive assessment, and purchase intention.

H2-Alternative: Negative emotion positively predicts cognitive assessment and purchase intention, and it will not influence perceived performance and perceived dividend.

Materials & Methods

Participants

Through an online invitation from a Taiwanese insurance company, 745 Taiwanese clients who held existing ULIP policies completed online questionnaires. To encourage participation in the online experiment, they were provided with an incentive—a chance to enter a drawing for 150 NTD coffee coupons. After excluding 176 participants who did not pass an attention check (i.e., those who did not respond with the designated response “strongly agree” to an attention check question), data from a total of 569 Taiwanese participants (Age range 23–76, $M_{\text{Age}} = 47.43$, $SD_{\text{age}} = 10.02$, 69% female, 31% male, 61% having a university degree) were included in the analysis.

Experimental Design

A randomized between-subjects design was implemented. Participants were randomly assigned to one of two conditions. The survey questionnaires were identical across the conditions; however, the image and text presented in the mood induction procedure were different in inducing positive and negative emotions, respectively.

The entire experiment was conducted on the online questionnaire platform Qualtrics. This research was approved by the Research Ethics Committee at National Taiwan University (NTU-REC-202011HS011) and

the whole survey procedure was performed in accordance with Research Ethics Committee's guidelines. All participants will fill out an informed consent form before proceeding with questionnaires. Participants were asked to report their current emotions in a questionnaire, then they filled out online questionnaires of demographic information. The questionnaires were followed by the mood induction procedure, and their emotion was measured immediately after the mood induction procedure using the same questionnaire as before. They were then asked to report their evaluation of the ULIP product profile, such as perceived performance, perceived dividend, and cognitive assessment. Finally, they participated in a hypothetical investment scenario, where they endowed an amount of 200,000 NTD and had to decide the amount of investment in the product.

Materials and Measures

Emotion Induction Procedure

Two images adapted from the Open Affective Standardized Image Set (OASIS)³⁴ were used to evoke the positive and negative emotions of the participants. OASIS is an open-access online database containing 900 images with empirical ratings on dimensions of valence and arousal. While inducing emotions through images is effective in an online environment³⁵, we selected six images with higher ratings of arousal as candidates to ensure the effectiveness of the emotion induction procedure. To determine the two images that perform the best in evoking positive and negative emotion, respectively, the images were then rated by 10 students at National Taiwan University. Through the initial screening process, we chose one image (Fig. 3) as the positive valence image and another (Fig. 4) as the negative valence image, and these two pictures present high arousal and decent reliability in the pilot study's result. We also used texts to induce positive or negative emotions, as they showed effectiveness in the online experimental environment³⁶. Texts inducing positive and negative emotions were equal in word count and had similar sentence structure in Chinese.

Perceived performance and dividend

A profile of ULIP was designed as the experiment material (Fig. 5). As ULIP contracts are complicated, we might achieve a more realistic investment environment by presenting more available information to the participants. However, a dilemma exists that their investment decisions can be biased due to information overload³⁷. After consulting an expert at the insurance institution, we chose to include two key product characteristics in the profile, the performance of a fund linked to the ULIP (presented as the rate of return at numeral time points) and the monthly dividend rate (set as 3.8%). The values of the key product characteristics were derived from actual data from ULIP and endorsed by experts. Participants were asked to evaluate the dividend and performance of the current ULIP using two questions on a six-point Likert scale, ranging from strongly disagree (1) to strongly agree (6).

Emotion

Sixteen items were selected from the Positive And Negative Affect Schedule (PANAS)³⁸ to assess participants' current emotions. Four items (Irritable, Alert, Ashamed, Attentive) in the original scale are excluded due to low validity, identified with factor analysis in a pilot study. An example of a positive emotion item was "enthusiastic," while a negative emotion item was "afraid". Participants provided ratings on a six-point Likert scale, ranging from strongly disagree (1) to strongly agree (6).

Cognitive Assessment

Five items modified from the Personal Involvement Inventory scale³⁹ and Perceptual Risk Factor measurements⁴⁰ were used to measure the cognitive assessment of the ULIP profile. Sample questions were "This unit-link insurance plan can meet my needs." and "Purchasing this unit-link insurance plan allows me to earn expected returns.". Participants rated statements on a six-point Likert scale from strongly disagree (1) to strongly agree (6).

Data Analysis

Data were analyzed using R (Version 4.1.2) and RStudio (2023.06.2). The comparison between the positive induction group and negative induction group was tested by an independent T-test, which was executed by the "stats" package in R. In addition, a path analysis model using the maximum likelihood estimation in R studio and "lavaan" package was performed. The standardized (β) path coefficients were reported so that the relative effect sizes on endogenous variables between variables can be compared by using the standardized regression weights.

Data Availability

The raw data and code of this study are uploaded to OSF (https://osf.io/thqgm/?view_only=d911c19d52be488b96f57b0ac2fe02c0). This study's design and its analysis were not preregistered. Question items used for the current study are listed in Appendix B.

Results

Descriptive Statistics and Correlation

Descriptive statistics and correlations are provided in Table 1. Manipulated positive emotions showed positive correlations with Perceived performance ($r = 0.20, p < .001$), Perceived dividend ($r = 0.11, p < .001$), Cognitive Assessment ($r = 0.28, p < .001$), and Willingness to pay ($r = 0.25, p < .001$). However, Manipulated negative emotions only showed negative correlations with perceived performance ($r = -0.12, p < .001$).

Table 1
Descriptive Statistics and Correlations of Study Variables

Variable	M	SD	1	2	3	4	5	6
1.Positive emotion after manipulation	3.90	0.87	-					
2.Negative emotion after manipulation	2.48	0.97	-0.40***	-				
3.Perceived performance	3.95	1.03	0.20***	-0.12***	-			
4.Perceived dividend	3.66	1.04	0.11***	-0.04	0.51***	-		
5. Cognitive Assessment	3.58	1.01	0.28***	-0.04	0.52***	0.51***	-	
6.Willingness to pay	3.36	1.04	0.25***	-0.01	0.53***	0.56***	0.80***	-

Note. N = 569. * $p < .05$; ** $p < .01$; *** $p < .001$.

Effectiveness of the Mood-Induction Procedure

As an improvement of the experimental design compared with a previous study that also examined the effect of emotions on decision-making⁴¹, we measured participants' initial levels of affect. Before emotion induction, the initial positive emotion ($t(567) = 0.23, p = .816$) and negative emotion ($t(567) = -0.95, p = .341$) were not different between the positive manipulation group and the negative manipulation group. After emotion induction, positive emotion in the positive manipulation group was stronger than that in the negative manipulation group ($t(567) = 4.59, p < .001$). Similarly, negative emotion in the negative manipulation group was higher than in the positive manipulation group ($t(567) = 3.66, p < .001$). These results indicate that the emotion induction did elicit emotional responses in the expected directions.

Fitness of the hypothetical path model

The hypothesized model was first tested by path analysis. Fit indices showed that our model had a good fit to the data [$\chi^2(22, N = 569) = 84.441, p < .05$; RMSEA = 0.071 (90% CI = [.055, .087]); CFI = .950; SRMR = 0.037].

Path analysis results

Keith (2019)⁴² proposed a rule of thumb for the interpretation of effect sizes in direct effects: Standardized parameter coefficients with values smaller than 0.05 are too small to be meaningful, values larger than 0.05 are small but meaningful, values larger than 0.10 are moderate, and values larger than 0.25 are large. The standardized parameter coefficients of each path are shown in Fig. 6 (refer to Appendix A for detailed information).

First, we examined the efficiency of emotion inducement. The positive manipulation group was coded in “1” and the negative manipulation group was coded in “-1”. After controlling the initial emotion state, emotion manipulation had a positive effect on positive emotion and a negative effect on negative emotion perceptively. Results revealed that both inducement groups elicited the targeted emotion successfully.

Second, we examined the mechanisms by which emotions influenced perception, cognitive assessment, and investment willingness to the ULIP. Results revealed that positive emotion directly enhanced cognitive assessment of ULIP. Moreover, participants after positive emotion induction perceived better performance and dividend of the ULIP, which then improved their cognitive assessment of the product. In comparison, negative emotions were also positively related to higher cognitive assessment. However, negative emotions did not influence the perception of ULIP.

To compare the prediction strength of positive and negative emotions on cognitive assessment, we set a null hypothesis that the coefficients of positive and negative emotions are not different from each other. The following regression comparison test in SEM shows that the above null hypothesis should be rejected and positive emotions have stronger predictive effects than negative emotions ($p = .001$, 95% CI=[.005, .055]). Eventually, either manipulated positive emotion or negative emotion would inspire people to buy ULIP through improved cognitive assessment. In addition, by considering the covariance between positive and negative emotions, we were able to examine how changes in one emotion corresponded to changes in the other, thus capturing the interplay between positive and negative emotional states.

Finally, to examine the framing effect induced by emotions, we used both positive and negative emotions to predict one item in the cognitive assessment: “Purchasing this unit-link insurance plan allows me to earn expected returns.”. The result showed that positive emotion improved the expectation of gain (Estimation = 0.36, $p = .001$). Nevertheless, negative emotion did not predict this item (Estimation = 0.08, $p = .119$). This finding suggests that the gain framing in ULIP may be influenced by positive emotion.

Discussion

Our model reveals that individuals exposed to both positive and negative emotional stimuli exhibit an enhanced cognitive evaluation of ULIP, thereby increasing their willingness to purchase this product. Furthermore, we observed that positive emotion exerts a stronger influence on cognitive evaluation than negative emotion. The acceptance of H1 and rejection of H1-alternative validate our predictions regarding the positive impact of positive emotion. Conversely, H2 was rejected and H2-alternative was accepted, which means that negative emotion positively predicts cognitive assessment and purchase intention while having no impact on perceived dividends and perceived performance. Our results demonstrate that when investors are in positive emotions, the decision-making pathway to purchase ULIP tends to regard it as an investment product, whereas when investors are in negative emotions, the decision-making pathway to purchase ULIP tends to regard it as an insurance product. This underscores

the complex interplay between emotions and decision-making during the investment process, with positive and negative emotions following distinct pathways.

Unlike traditional investment and insurance products, ULIP is a rising product combining investment and insurance features. Customers may perceive one of these two aspects more prominently during the purchasing process due to the inherent conflicts between insurance and investment characteristics. Drawing from the AIM model and framing effect, we infer that different emotions may prime customers to perceive distinct perspectives and information regarding ULIPs. Individuals experiencing positive emotions tend to perceive a gain frame and maintain an optimistic outlook on the future of the investment market^{14,15}. Conversely, negative emotions may trigger expectations of potential future crises, prompting individuals to seek protection by purchasing insurance to mitigate these negative feelings, which is a loss frame²¹⁻²³.

Therefore, we posit that consumers experiencing positive emotions are more likely to perceive ULIPs as investment products under the gain frame, consistent with previous literature highlighting the positive correlation between positive emotions and investment willingness^{14,15}. Conversely, consumers experiencing negative emotions are more inclined to view ULIPs as insurance products under the loss frame, aligning with previous research demonstrating that negative emotions facilitate the purchase of insurance products. In essence, when deciding whether to purchase ULIPs, the induction of either positive or negative emotions can enhance cognitive evaluations and purchase intentions through distinct pathways.

AIM also provides insights into why positive and negative emotions within our study framework led to different decision-making pathways. According to the affect-priming process, attention is a variable susceptible to the influence of positive and negative emotions, subsequently affecting information extraction and processing. Fredrickson's (2001)⁴³ broaden-and-build theory suggests that positive emotions broaden an individual's momentary thoughts and actions, fostering the expansion of resources in various aspects such as creativity, social behavior, physical activity, curiosity, and more. Conversely, negative emotions narrow an individual's mindset. Research indicates that negative emotions reduce attentional scope, while positive emotions broaden it⁴⁴. During judgment tasks, previous studies have shown that positive and negative emotions prime individuals to focus on global information and local features, respectively⁴⁵. Additionally, an eye-tracking study found that, compared with the control group, positive emotion manipulation caused individuals to pay more attention to peripheral stimuli⁴⁶. Therefore, in our context of reading the ULIP profile, positive emotions may direct participants' attention to additional textual and numerical information related to investment, such as performance and dividend payments. On the other hand, individuals experiencing negative emotions may overlook these detailed pieces of information.

This inference is consistent with our ULIP purchasing model. Within our model framework, individuals experiencing positive emotions have a higher cognitive evaluation and also perceive a higher performance and dividend. At the same time, individuals experiencing negative emotions will only have a

higher cognitive evaluation without perceiving the difference in performance and dividend. Instead, they directly perceive a need for ULIPs, leading to an increased willingness to purchase.

Drawing from the broaden-and-build theory and previous research on attention, we posit that positive emotions allocate more cognitive resources to process information on the ULIP profile, rendering it more readable. Consequently, profitable information gains prominence during the information processing stage. Additionally, research has shown that increased readability in disclosures elicits stronger responses from small investors, which then results in more positive valuation judgments when the news is favorable and more negative judgments when the news is unfavorable⁴⁷. Taken together, we deduce that positive emotions enhance the readability of ULIP information and amplify the impact of favorable information during the information processing stage.

In conclusion, our findings indicate that positive emotions prompt investors to perceive ULIPs as investment products, pay attention to investment information, make optimistic predictions about investment, and therefore desire to buy ULIPs. On the other hand, negative emotions prompt investors to perceive ULIPs as insurance products, triggering a feeling of urgency and a desire to purchase. When investors decide whether to buy ULIPs, being emotionally aroused, either positively or negatively, enhances their willingness to make the purchase, but with distinct underlying decision-making processes.

Contributions

Our findings have several practical implications for marketing and communication strategies. Marketers targeting potential ULIP customers should take into account the emotional state of individuals⁴⁸. By understanding how emotions influence cognitive evaluations and decision-making pathways, marketers can tailor their messages to elicit specific emotions that align with the desired perception of ULIPs. For instance, when targeting individuals with positive emotions, highlighting the investment features and potential returns of ULIPs may be more effective. Conversely, when targeting individuals with negative emotions, emphasizing the insurance benefits and risk mitigation aspects of ULIPs may be more persuasive.

Product designers and managers should consider the emotional context in which ULIPs are presented⁴⁸. By incorporating elements that resonate with positive emotions, designers can enhance the perceived investment value of ULIPs. Similarly, incorporating elements that address negative emotions can strengthen the perceived insurance value of ULIPs. This careful product design and positioning can help customers better understand and appreciate the unique features of ULIPs.

In addition, the investment and insurance industry should recognize the influence of emotions on customers' perceptions of ULIPs⁴⁹. This understanding can inform the development of customer-centric approaches⁵⁰ in product offerings, sales processes, and customer service interactions. Financial advisors and agents can adapt their communication strategies based on the emotional state of

customers, tailoring their advice and recommendations to address customers' emotional concerns and priorities. By incorporating emotional intelligence into industry practices, professionals can enhance customer engagement and satisfaction⁵¹.

On the other side, by providing comprehensive and balanced education about ULIPs, individuals can make informed decisions based on their financial goals and emotional states. Customer education and financial literacy programs play a vital role in improving individuals' understanding of ULIPs and their emotional impact. They can help individuals navigate the complexities of ULIPs, enabling them to recognize and manage the emotional biases that may influence their decision-making. These programs contribute to more confident and rational decision-making regarding ULIPs^{52,53}.

Regulatory frameworks should also consider the emotional dimensions of consumer decision-making regarding ULIPs. Policymakers can incorporate emotional aspects into consumer protection measures and disclosure requirements. This ensures that individuals are equipped with the necessary information and safeguards to make well-informed decisions regarding ULIPs, considering their emotional biases and vulnerabilities. Such regulations can contribute to a more transparent and fair marketplace for ULIPs, promoting consumer confidence and trust.

Limitations and Future Directions

Some limitations to our studies should be acknowledged. In our study, we propose that consumers may perceive ULIPs as either leaning toward investment products or as a form of insurance. Positive and negative emotions may prompt individuals to pay attention to different aspects of ULIPs, emphasizing either the investment or insurance features, respectively, thereby triggering distinct cognitive processes⁵⁴. However, we did not directly measure consumers' subjective perceptions of whether ULIPs resemble investment products or insurance. Therefore, this argument serves as an explanatory discussion and more evidence awaits further investigation in future research to explore this aspect more comprehensively.

Future research can utilize eye-tracking technology to gain deeper insights into consumers' attentional patterns when evaluating ULIPs⁵⁵. Eye tracking provides valuable information on participants' gaze patterns, fixation duration, and which aspects of ULIPs attract the most attention. Researchers can investigate whether individuals focus more on investment-related information or insurance-related information in ULIP descriptions. In our current study, it's important to note that we did not specifically differentiate between investment-related and insurance-related information within ULIP descriptions. Instead, our research is based on the composition of ULIPs on the market. We treated ULIPs as hybrid financial products with both insurance and investment characteristics, encompassing aspects such as dividend payouts and performance. Future studies could build upon this foundation to further explore and distinguish the attentional patterns related to these distinct aspects, thereby providing a more comprehensive understanding of how individuals perceive ULIPs. Researchers can also examine if positive emotions lead to increased attention towards investment features and negative emotions draw

attention towards insurance features. Eye-tracking data can also be used to compare groups of participants with varying levels of investment knowledge or prior experiences with insurance products, revealing divergent attentional patterns and subjective perceptions of ULIPs. By employing eye-tracking technology, future studies can validate and expand upon the proposed theoretical framework.

Also, we utilized self-reported measures to assess consumers' willingness to purchase, which may not entirely capture their actual investment decisions or behaviors in real-world contexts^{56,57}. Hence, the translation of reported intentions into behaviors remains uncertain. For a more profound understanding of consumer behavior and decision-making concerning ULIPs, future research could integrate behavioral measures. This may involve tracking the actual monetary investments made in these products^{53,58} to provide a more comprehensive insight into consumer actions and their relationship with ULIPs.

Conclusion

Our proposed model demonstrates that both positive and negative emotions enhance cognitive evaluations and investment willingness toward ULIPs. Drawing on the AIM theory, we infer that positive and negative emotions induce different framings (gain and lost) and expand or restrict attention, respectively, which in turn leads consumers to interpret ULIPs differently. Under positive emotions, ULIPs are perceived as investment products, while under negative emotions, they are perceived as insurance products. These elicited positive and negative emotions independently facilitate investment willingness towards the perceived product type. Therefore, the emotion elicited promotes the purchase of ULIPs through different pathways.

Moreover, our research provides practical insights for various stakeholders involved in ULIPs, including marketers, product designers, educators, industry professionals, and policymakers. By considering the influence of emotions on cognitive evaluations and decision-making, these stakeholders can develop strategies and initiatives that enhance customer engagement, satisfaction, and informed decision-making in the context of ULIPs.

Declarations

Author Contribution

Peng-Yu Zeng, Hsuan-Yu Liu, Li-Yu Chen, and Su-Ling Yeh designed the experiments. Peng-Yu Zeng, Hsuan-Yu Liu, and Li-Yu Chen performed the experiments and analyzed the data. Peng-Yu Zeng, Hsuan-Yu Liu, and Li-Yu Chen wrote the paper. Su-Ling Yeh critically revised the manuscript. All authors reviewed the final version of the manuscript.

Acknowledgement

We thank Chien-Yu Tsai, Sheng-Mou Lin, Yu-Hua Huang, Chen-Wei Huang, and Ze-Li Lin for their assistance in preparing stimuli and questionnaires in the early stage of the research.

Data Availability

The raw data and code of this study are uploaded to OSF (https://osf.io/thqgm/?view_only=d911c19d52be488b96f57b0ac2fe02c0). This study's design and its analysis were not preregistered. Question items used for the current study are listed in Appendix B.

References

1. Gantenbein, M. & Mata, M. A. Swiss Annuities and Life Insurance: Secure Returns, Asset Protection, and Privacy. (John Wiley & Sons, 2008).
2. Mandal, S. Unit linked Insurance Plans and their applications in India. (Thesis, 2016).
3. Akula, R. & Kanchu, T. Growth of Ulip Policies in Life Insurance Sector - A Comparative Study of Traditional and Ulip Policies. *Indian Journal of Commerce and Management Studies* 2, 190–202 (2011).
4. Kahneman, D. & Tversky, A. Prospect Theory: An Analysis of Decision Under Risk. in *World Scientific handbook in financial economic series* 99–127 (2013).
5. Loewenstein, G. F., Weber, E. U., Hsee, C. K. & Welch, N. Risk as feelings. *Psychological Bulletin* 127, 267–286 (2001).
6. Cabanac, M. What is emotion? *Behavioural Processes* 60, 69–83 (2002).
7. Forgas, J. P. Mood and judgment: The affect infusion model (AIM). *Psychological Bulletin* 117, 39–66 (1995).
8. Bower, G. H. Mood and memory. *American Psychologist* 36, 129–148 (1981).
9. Deldin, P. J. & Levin, I. P. The effect of mood induction in a risky decision-making task. *Bull. Psychon. Soc.* 24, 4–6 (1986).
10. Breaban, A. & Noussair, C. N. Emotional State and Market Behavior. *Review of Finance* 22, 279–309 (2018).
11. Chou, K.-L., Lee, T. M. C. & Ho, A. H. Y. Does mood state change risk taking tendency in older adults? *Psychology and Aging* 22, 310–318 (2007).
12. Wigfield, A. & Eccles, J. S. Expectancy–Value Theory of Achievement Motivation. *Contemporary Educational Psychology* 25, 68–81 (2000).
13. Carver, C. S. & Scheier, M. F. Optimism, pessimism, and self-regulation. (2001).
14. Snyder, C. R. Hope Theory: Rainbows in the Mind. *Psychological Inquiry* 13, 249–275 (2002).
15. Knutson, B., Taylor, J., Kaufman, M., Peterson, R. & Glover, G. Distributed Neural Representation of Expected Value. *J. Neurosci.* 25, 4806–4812 (2005).

16. Angelini, V. & Cavapozzi, D. Dispositional optimism and stock investments. *Journal of Economic Psychology* 59, 113–128 (2017).
17. Eisenberg, A. E., Baron, J. & Seligman, M. E. Individual differences in risk aversion and anxiety. *Psychological Bulletin* 87, 245–251 (1998).
18. Maner, J. K. et al. Dispositional anxiety and risk-avoidant decision-making. *Personality and Individual Differences* 42, 665–675 (2007).
19. Lindquist, K. A. & Barrett, L. F. Constructing Emotion: The Experience of Fear as a Conceptual Act. *Psychol Sci* 19, 898–903 (2008).
20. Hsee, C. K. & Kunreuther, H. C. The Affection Effect in Insurance Decisions. *Journal of Risk and Uncertainty* 20, 141–159 (2000).
21. Shen, Y., Yang, Q. & Zhao, C. Food safety risks and defensive behavior: Evidence from health insurance. *Finance Research Letters* 58, 104027 (2023).
22. Brighetti, G., Lucarelli, C. & Marinelli, N. Do emotions affect insurance demand? *Review of Behavioral Finance* 6, 136–154 (2014).
23. Kunreuther, H. & Pauly, M. V. Dynamic Insurance Decision-Making for Rare Events: The Role of Emotions. *Geneva Pap Risk Insur Issues Pract* 43, 335–355 (2018).
24. Yan, J., Kniffin, K. M., Kunreuther, H. C. & Schulze, W. D. The roles of reason and emotion in private and public responses to terrorism. *Journal of Economic Behavior & Organization* 180, 778–796 (2020).
25. Jaspersen, J. G. & Aseervatham, V. The Influence of Affect on Heuristic Thinking in Insurance Demand. *Journal of Risk and Insurance* 84, 239–266 (2017).
26. Hogarth, R. M. & Kunreuther, H. Risk, ambiguity, and insurance. *J Risk Uncertainty* 2, 5–35 (1989).
27. Koufopoulos, K. & Kozhan, R. Welfare-improving ambiguity in insurance markets with asymmetric information. *Journal of Economic Theory* 151, 551–560 (2014).
28. Johnson, E. J., Hershey, J., Meszaros, J. & Kunreuther, H. Framing, probability distortions, and insurance decisions. *J Risk Uncertainty* 7, 35–51 (1993).
29. Mittal, V. & Ross, W. T. The Impact of Positive and Negative Affect and Issue Framing on Issue Interpretation and Risk Taking. *Organizational Behavior and Human Decision Processes* 76, 298–324 (1998).
30. Nabi, R. L. et al. Can Emotions Capture the Elusive Gain-Loss Framing Effect? A Meta-Analysis. *Communication Research* 47, 1107–1130 (2020).
31. Charles, A. & Kasilingam, R. Do Investor's Emotions Determine Their Investment Decisions? *Drishtikon: A Management Journal* 6, (2015).
32. Gupta, A. Unit Linked Insurance Products (ULIPs) - Insurance or Investment? *Procedia - Social and Behavioral Sciences* 37, 67–85 (2012).
33. Lie, E. Operating performance following dividend decreases and omissions. *Journal of Corporate Finance* 12, 27–53 (2005).

34. Kurdi, B., Lozano, S. & Banaji, M. R. Introducing the Open Affective Standardized Image Set (OASIS). *Behav Res* 49, 457–470 (2017).
35. van Boxel, K., Decke, P., Nolte, S. & Schneider, J. C. Positivity and Nature: How Images Increase Investment. (2021).
36. Verheyen, C. & Göritz, A. S. Plain Texts as an Online Mood-Induction Procedure. *Social Psychology* 40, 6–15 (2009).
37. Agnew, J. R. & Szykman, L. R. Asset Allocation and Information Overload: The Influence of Information Display, Asset Choice, and Investor Experience. *Journal of Behavioral Finance* 6, 57–70 (2005).
38. Watson, D., Clark, L. A. & Tellegen, A. Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology* 54, 1063–1070 (1988).
39. Zaichkowsky, J. L. The Personal Involvement Inventory: Reduction, Revision, and Application to Advertising. *Journal of Advertising* 23, 59–70 (1994).
40. Chang Y.-C. The Study of Money Attitude, Perceived Risk and Consumer Purchase Intention on Investment Linked Insurance Products. (2013).
41. Vanwalleghe, D. & Mirowska, A. The investor that could and would: The effect of proactive personality on sustainable investment choice. *Journal of Behavioral and Experimental Finance* 26, 100313 (2020).
42. Keith, T. Z. Multiple Regression and beyond: An Introduction to Multiple Regression and Structural Equation Modeling. (Routledge, 2019).
43. Fredrickson, B. L. The role of positive emotions in positive psychology: The broaden-and-build theory of positive emotions. *American Psychologist* 56, 218–226 (2001).
44. Huntsinger, J. R. Does Emotion Directly Tune the Scope of Attention? *Curr Dir Psychol Sci* 22, 265–270 (2013).
45. Fredrickson, B. L. & Branigan, C. Positive emotions broaden the scope of attention and thought-action repertoires. *Cognition and Emotion* 19, 313–332 (2005).
46. Wadlinger, H. A. & Isaacowitz, D. M. Positive mood broadens visual attention to positive stimuli. *Motiv Emot* 30, 87–99 (2006).
47. Rennekamp, K. Processing Fluency and Investors' Reactions to Disclosure Readability. *Journal of Accounting Research* 50, 1319–1354 (2012).
48. Gobe, M. Emotional Branding: The New Paradigm for Connecting Brands to People. (Simon and Schuster, 2010).
49. Lee, M., Jeong, M. & Lee, J. Roles of negative emotions in customers' perceived helpfulness of hotel reviews on a user-generated review website: A text mining approach. *International Journal of Contemporary Hospitality Management* 29, 762–783 (2017).

50. Bolton, M. Customer centric business processing. International Journal of Productivity and Performance Management 53, 44–51 (2004).

51. Kim, S.-H. The Effect of Emotional Intelligence on Salesperson’s Behavior and Customers’ Perceived Service Quality. Proceedings of the Korean Distribution Association Conference 127–158 (2007).

52. Lusardi, A. Financial Literacy: An Essential Tool for Informed Consumer Choice? (2008).

53. Lusardi, A. & Mitchell, O. S. Financial Literacy and Planning: Implications for Retirement Wellbeing. Working Paper at <https://doi.org/10.3386/w17078> (2011).

54. Liew, T. W. & Tan, S.-M. The Effects of Positive and Negative Mood on Cognition and Motivation in Multimedia Learning Environment. Journal of Educational Technology & Society 19, 104–115 (2016).

55. Holmqvist, K., Andersson, R., Dewhurst, R., Jarodzka, H. & Van de Weijer, J. Eye Tracking: A Comprehensive Guide to Methods and Measures. (oup Oxford, 2011).

56. Schwarz, N. Why researchers should think “real-time”: A cognitive rationale. Handbook of research methods for studying daily life 22, (2012).

57. Tourangeau, R. & Yan, T. Sensitive questions in surveys. Psychological Bulletin 133, 859–883 (2007).

58. Odean, T. Volume, Volatility, Price, and Profit When All Traders are Above Average. (1998).

Figures

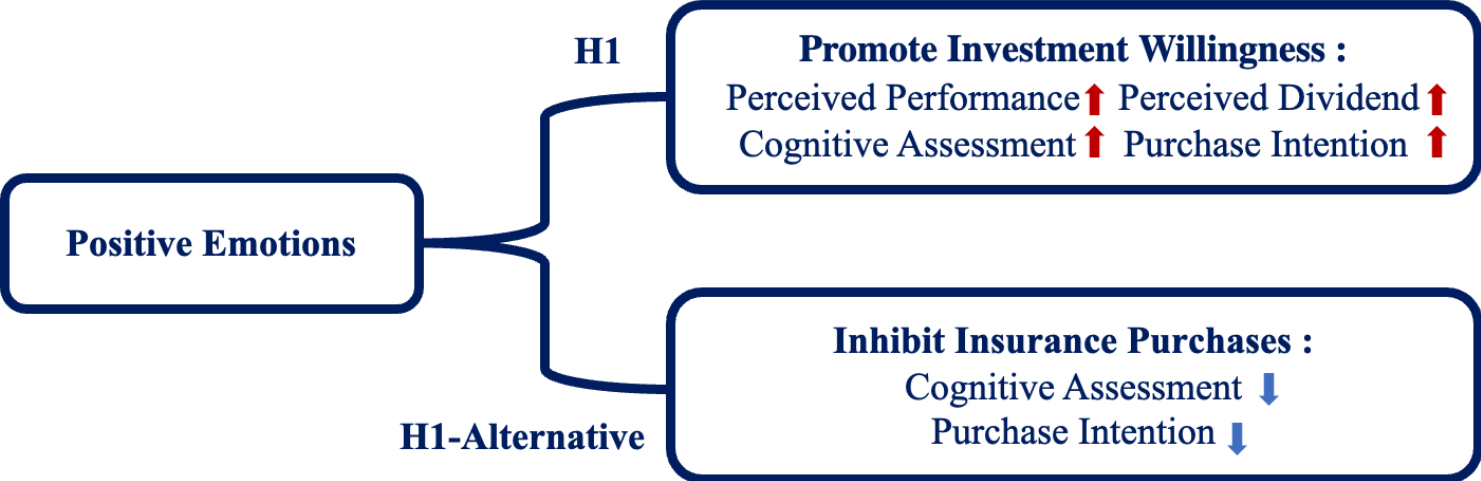


Figure 1

Hypothesized Influence of Positive Emotions on ULIP Product Purchases products

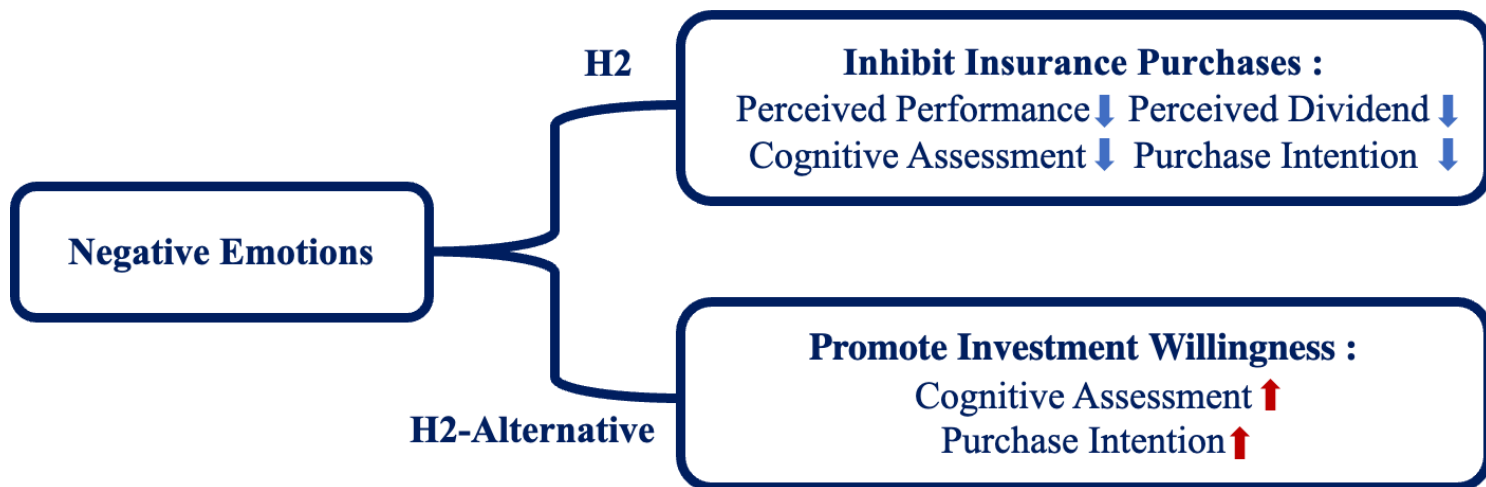


Figure 2

Hypothesized Influence of Negative Emotions on ULIP Product Purchases



My primary goal is to achieve financial independence. Besides my work income, I also generate money through financial investments. I feel fortunate to have the opportunity for early retirement, travel, and enjoying life.

Figure 3

The material used in the emotion induction procedure for the positive emotion group (text translated from Chinese).



My primary goal is to maintain a stable income. Besides fixed expenses, I've incurred significant costs due to health decline. I feel concerned about potential additional expenses, decreased income, and the possibility of being unwell and financially strained in case of an accident.

Figure 4

The material used in the mood induction procedure for the negative emotion group (text translated from Chinese).

Fu Lu Insurance Unit-Link Insurance Plan



Performance

Performance : Return on investment of the fund linked to this unit-linked insurance in the past five years.

Fund Name	Internal Rate of Return							
	1st month %	3rd month %	6th month %	This Year	1st Year %	2nd Year %	3rd Year %	5th Year %
This Fund	-1.84%	-0.78%	-2.63%	-2.57%	1.71%	22.71%	41.82%	53.79%
Average of other similar funds	-1.41%	-0.72%	-1.77%	-1.91%	2.54%	14.38%	33.08%	45.23%



Dividend

Dividend unit per month	Internal Rate of Dividend
1.02	5%



Fu Lu Insurance

Fu Lu Financial Holding

Figure 5

The ULIP profile contains details regarding performance and dividends (translation from Chinese). The company names and logos mentioned above are purely fictitious and were generated solely for research purposes only.

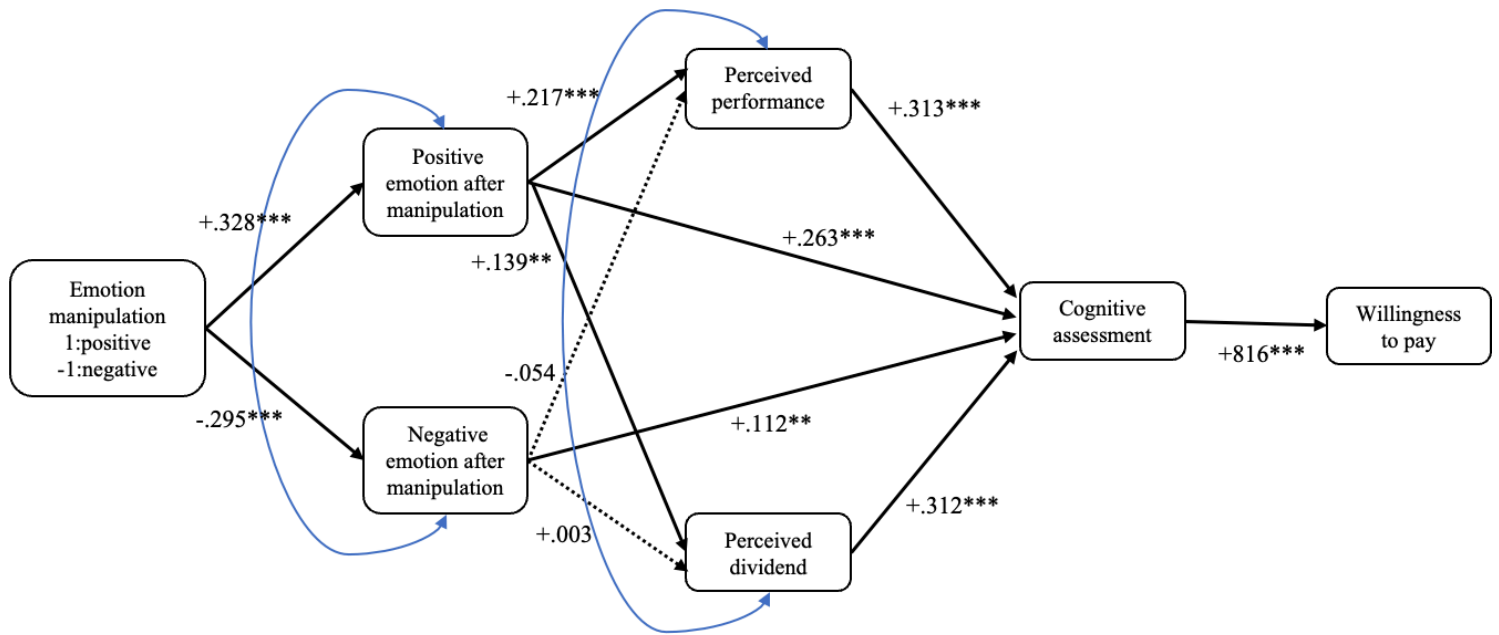


Figure 6

$N = 569$. Path diagram for the ULIP purchasing model. The solid lines and the dotted lines represent that the regression coefficients were significant ($p < .05$) and not significant ($p > .05$) respectively. In the block of emotion manipulation, the group coding of positive emotion manipulation is 1, and the group coding of negative emotion manipulation is -1. The variables connected by the blue line (dark gray) were controlled as a covariance.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [SupplementaryMaterial.docx](#)